

Between Law and Numbers: Child Marriage and Adolescent Pregnancy in Brazil

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Abstract

Child marriage –defined as any formal or informal union in which at least one party is under minimum legal age –remains a significant social and economic challenge in Brazil. Until 2019, children under 16 were allowed to marry in cases of pregnancy or to avoid criminal penalties, but after Law 13,911 was enacted, it is illegal for children under 16 years old to be married. Although, in 2022, civil registries recorded 260 cases with individuals younger than 16, which represents 1.8% of child marriages in this period. This paper provides a comprehensive descriptive analysis and model the discrete choice of child marriage in Brazil over the period from 2003 to 2022, combining complementary data sources: Civil Registry Statistics and the National Continuous Household Sample Survey (PNADC).

We document a substantial decline in child marriage over the period. Unions involving minors declined by approximately 72% per 10,000 inhabitants. We further show that the observed decline in formal marriages is partly driven by a shift toward informal unions, which are not captured in civil registry data. By comparing administrative and survey sources, we estimate that official records understate the prevalence of unions involving minors by 75% to 83%. These findings present measurement issues in understanding the persistence of early unions and the weak enforcement of legal restrictions. This paper represents the first stage of a broader research agenda that aim to evaluate the causal impact of the legal reform on child marriage, adolescent fertility, and related socioeconomic outcomes.

Keywords: Child Marriage; Informal Unions; Underreporting

JEL codes: J12, J13, J16, I38, K36

1 Introduction

Child marriage not only violates the rights of children and adolescents, but also exacerbates violence, limits educational and economic opportunities, increases early pregnancies, and raises maternal and infant mortality rates. Additionally, it contributes to intergenerational poverty and other social challenges, making it a multidimensional issue (Field and Ambrus, 2008); (Cardoso et al., 2022); (United Nations Children’s Fund (UNICEF), 2021).

In Brazil, legislation sets 18 as the minimum legal age for marriage. However, individuals aged 16 to 18 may marry with parental consent, and until 2019, minors under 18 could also marry under specific circumstances, such as pregnancy or to avoid criminal penalties, provided judicial authorization was granted. Law 13,811 of March 12, 2019 amended Article 1,520 of the 2002 Civil Code, eliminating these exceptions. Currently, marriage is strictly prohibited for individuals under 16 and allowed for those aged 16 and above only with parental consent (Brasil, 2019). Prior to this reform, perpetrators of statutory rape could avoid punishment by marrying the victim if she was under 18 and parental consent was granted (Brasil, 2009).

Evidence from civil registry data¹ indicates a decline of approximately 72% in child marriages between 2003 and 2022, broadly consistent with trends observed in other age groups. Nevertheless, in 2022 alone, 15,172 formal unions involving individuals under 18 were registered, which corresponds to 6,1 marriages per ten thousand minors in that year. Despite the legal prohibition, civil registry data still report marriages involving individuals under 16 – 260 cases in 2022 alone, which corresponds to 1,8% of child marriages registered that year.

These figures suggest that, although declining, child marriage remains a relevant issue in civil registry, which does not account for the informal unions formed of children. Informal arrangements are especially prevalent among minors, as a result, the true magnitude of the phenomenon is likely significantly underestimated.

Although child marriage have declined over time, important structural mechanisms persist. In particular, early unions remain highly gendered, disproportionately affecting girls, who also tend to enter such unions at younger ages. Moreover, while official statistics indicate a substantial reduction in marriages involving minors, complementary evidence points to a growing role of informal unions, which are not captured in civil registry data.

Despite these patterns, child marriage in Brazil have received limited attention in the economics literature. There is a lack of comprehensive evidence on the most vulnerable populations, regional heterogeneity, and the broader consequences for human capital accumula-

¹Civil Registry Statistics from Instituto Brasileiro de Geografia e Estatística (IBGE)

tion, labor market outcomes, and fertility behavior. There is scarce quantitative evidence on the effectiveness of policy interventions aimed at delaying marriage among minors.

Therefore, this study examines the persistence of child marriage in Brazil. By combining survey and administrative data, this study shows that official records significantly underestimate the prevalence of unions involving minors, with implied underreporting rates ranging from approximately 75% to 83%. However, understanding these phenomena requires going beyond descriptive statistics and analyzing the institutional, legal context and economic measures that shapes incentives, constraints, and social norms.

Accordingly, this paper contributes not only by documenting empirical patterns, but also by systematizing and consolidating the legal framework governing child marriage in Brazil. This institutional perspective is essential for interpreting the persistence of these phenomena, as well as the potential effectiveness of recent legal reforms.

These contributions are closely related to the economics literature on the effectiveness of legal reforms. [McGavock \(2021\)](#) exploit a reform in Ethiopia that raised the minimum legal age of marriage and find significant delays in early marriage, although effects vary across groups depending on social norms. More broadly, [Collin and Talbot \(2017\)](#) show that enforcement of minimum marriage age laws varies substantially across countries, and in many cases such laws are weakly enforced and have limited effects.

Consistent with this, [Wilson \(2022\)](#) document that increasing the legal age of marriage in several low- and middle-income countries raises the age at marriage and first birth, and improves labor market outcomes, although effects are weaker in rural areas. In contrast, [Batyra and Pesando \(2021\)](#) find that reductions in the legal marriage age had limited or inconsistent effects across countries, emphasizing the role of enforcement and social norms. From a different perspective, [Le, Nguyen, and Ibuka \(2025\)](#) show that education policies can indirectly delay marriage and fertility by increasing schooling, while [Buchmann et al. \(2023\)](#) find that financial incentives can be effective in postponing marriage, whereas attempts to directly shift social norms are less successful.

Taken together, this literature suggests that legal reforms alone may not be sufficient to substantially alter marriage behaviors, particularly in contexts where enforcement is weak and informal institutions are strong. This reinforces the importance of assessing the effectiveness of recent legal changes in Brazil and the mechanisms that maintain the choice of child marriage.

This paper contributes to this gap in four main ways. First, it provides a comprehensive descriptive analysis of child marriage in Brazil. Second, it highlights the central role of informal unions and the extent of underreporting, which are crucial for accurately measuring this phenomenon. Third, it systematizes the institutional and legal framework governing

child marriage in Brazil, providing a foundation for evaluating policy effectiveness. Fourth, it design the mechanisms underlying child marriage in a discrete choice model, where we can investigate which aspects are more relevant to sustain this equilibrium.

This paper represents the first, descriptive stage of a broader research agenda. Building on the patterns documented here, the next step of the project is to conduct a causal impact evaluation of Law 13.811/2019. By exploiting the reform as an exogenous institutional change, the objective is to estimate its effects on both formal and informal child marriage, as well as on related outcomes such as adolescent fertility, schooling, and labor market participation.

The remainder of the paper is organized as follows. Section 2 reviews the relevant literature and describes the institutional background. Section 3 discuss a model of discrete choice of child marriage in formal and informal margins. Section 4 presents the data. Section 6 reports descriptive results, presents equilibrium parameters before the law change and explore counterfactual policies in income, legal implications and social norms. Section 7 concludes and discusses paths for future research.

2 Related Literature and Institutional Background

We first understand the economic aspects of child marriage described in the literature and later we discuss the qualitative aspects of child marriage in Brazil.

2.1 The Economics of Child Marriage

The economics of child marriage is explored mostly through its outcomes in education, welfare and public health, as well as its driving mechanisms. We organize the relevant literature into three themes: consequences of early marriage, determinants and equilibrium mechanisms, and policy evaluation.

2.1.1 Consequences of Early Marriage

The first impact of early marriage is the children’s educational level, as they leave the schooling system to join household economic activities. A foundational contribution comes from [Field and Ambrus \(2008\)](#), who study rural Bangladesh using the timing of menarche as an instrument for age at first marriage. They find that one year of postponement of marriage between ages 11 and 16 increases schooling by 0.22 years and adult literacy by 5.6 percent. In this sense, [J. Hicks and D. L. Hicks \(2019\)](#) extends this evidence to western Kenya, confirming that delayed marriage improves educational attainment and perceptions

of health and well being. [C. Wang and L. Wang \(2017\)](#) further show that marriage delay raises wages for both men and women in the United States, with the effect operating almost entirely through greater educational attainment.

Beyond education, early marriage impacts directly the minor’s wealth, security and health, as well as other negative impacts for future generations of these marriages. [Dahl \(2010\)](#) uses variation in U.S. state-level marriage, schooling, and labor laws as instruments and finds that women who marry young are 31 percentage points more likely to live in poverty in adulthood. [Chari et al. \(2017\)](#) document inter generational consequences using Indian data: a one-year delay in marriage increases the probability that a child completes required vaccinations by 4.6 percent and raises school enrollment by 3.1 percent, providing evidence that the costs of child marriage extend well beyond the woman herself. Using sister fixed effects and menarche as an instrument in Bangladeshi data, [Asadullah and Zaki Wahhaj \(2019\)](#) show that early marriage increases agreement with traditional gender norms and reduces women’s exposure to non-traditional social networks, identifying an additional pathway through which early marriage perpetuates gender inequality across generations. Women married before age 18 face approximately 50 percent higher odds of experiencing physical or sexual intimate partner violence compared with women who married as adults ([Kidman, 2017](#)).

2.1.2 Determinants and Equilibrium Mechanisms

A second strand of the literature asks why child marriage persists as an equilibrium outcome. [Wahhaj \(2015\)](#) develops an overlapping generations model in which a girl’s “purity” is only noisily observed before marriage, so that remaining on the marriage market longer signals lower quality. In equilibrium, older brides face a reputational penalty and must pay higher dowries, generating a self-reinforcing cycle of early marriage. Crucially, the model predicts that policy impacts estimated on the first exposed cohort substantially underestimate long-run effectiveness, because they ignore the gradual improvement in the reputation of older women as the norm shifts over time.

[Corno, Hildebrandt, and Voena \(2020\)](#) shift attention to the role of aggregate economic shocks. Using rainfall variation across Sub-Saharan African countries and India, they show that drought increases the hazard of child marriage by 3 percent in Sub-Saharan Africa, where bride price prevails, but decreases it by 4 percent in India, where dowry is the norm. This sign reversal reveals that cultural institutions governing marriage payments mediate how income shocks translate into marriage timing. [Ashraf et al. \(2020\)](#) complement this finding through the Indonesian school construction program, showing that educational investments increase female schooling only in ethnic groups that practice bride price, because more educated

daughters command a higher price.

Therefore, income and social norms are both main drivers of child marriage, but the prevalence of each one depends on the type of income transferred and the cultural aspects of marriage and women’s sexuality.

2.1.3 Policy Evaluation

A third body of work evaluates the effectiveness of specific interventions. [Bharadwaj \(2015\)](#) finds that raising the cost of marriage in the 1957 amendment to Mississippi’s marriage law reduced marriages per 1,000 people by 75 percent, decreased crude birth rates by 2–6 percent, and raised school enrollment by 3 percent. [Wilson \(2022\)](#) scales this analysis to low- and middle-income countries, comparing subnational regions with high and low pre-ban marriage rates across countries that banned child marriage between 1995 and 2012. Raising the legal age to 18 increased the age at first birth and employment rates; in urban areas it also reduced child marriage and raised educational attainment, though effects on rural areas were more muted, consistent with weaker enforcement.

[Duflo, Dupas, and Kremer \(2015\)](#) test the interaction between education subsidies and an HIV prevention curriculum emphasizing abstinence until marriage. Education subsidies alone reduced dropout and delayed fertility; the abstinence curriculum alone did not reduce pregnancies or STIs and inadvertently shifted pregnancies toward marriage; the joint program reduced STI rates but was less effective on fertility than the subsidy alone. [Chow and Vivalt \(2022\)](#) evaluate an intervention in Ethiopia combining information campaigns with economic incentives, finding a 4–7 percentage point reduction in the probability of girls aged 8–17 being married, though the program triggered polarization in community beliefs and did not improve schooling.

Two papers raise important identification concerns. [Collin and Talbot \(2017\)](#) finds that only a small share of countries show credible enforcement, and that the introduction of more ambitious laws often coincides with weaker enforcement. [Blank, Charles, and Sallee \(2009\)](#) compares U.S. Vital Statistics administrative records with retrospective Census data, they show that estimates of the effect of marriage age laws differ dramatically by source, with administrative data capturing compliance and evasion behavior that retrospective surveys miss.

2.2 Child Marriage in Brazil

Brazil is the fourth country in the world with the highest absolute number of women married in childhood ([Margarido, 2024](#)). Despite the enactment of Law 13,811/2019 – which

eliminated all exceptions to marriage below age 16 – child marriage persists. The 2021 Civil Registry recorded 16,572 formal marriages involving minors, equivalent to 45.4 girls formally married per day (Margarido, 2024).

2.2.1 Prevalence and Epidemiological Evidence

Using nationally representative data from the 2013 National Health Survey, Cardoso et al. (2022) estimate the prevalence of marriage or union before age 18 at 3.9 percent. They show that being female, Black or Brown, residing in the North or Northeast, having lower educational attainment, and lower family income are all independently associated with higher prevalence.

Pregnancy among very young adolescents provides a complementary indicator of early union. Pinto et al. (2024) analyze births from mothers aged 10 to 14 in Brazil between 2011 and 2021. They find significant geographic clustering concentrated in the North and Center-West regions. Among these very young mothers, 73.6 percent were Black or Brown and 21.1 percent declared themselves married or in a stable union.

2.2.2 Policy Evaluation: The Bolsa Família Program

A contribution to the Brazilian policy literature evaluates the Bolsa Família Program (PBF), Brazil’s main conditional cash transfer. Vasconcelos and Griebeler (2017) find that PBF reduces the probability of child marriage among girls in extreme poverty, contradicting the hypothesis that the benefit inadvertently increases girls’ value as “brides”. Vasconcelos and Griebeler (2023) finds that the protective effect extends to girls in general poverty – beyond extreme poverty – and is most pronounced for girls aged 16 to 18. For the youngest cohort, ages 14 to 15, the effect is not statistically significant, suggesting that non-economic factors dominate in that age range.

2.2.3 Qualitative Dimensions

The qualitative literature reveals the cultural and psychological mechanisms sustaining child marriage in Brazil. Taylor and Fonseca (2015) studies main drivers of child marriage in Pará and Maranhão, they find that child marriage is driven by the family’s desire to control female sexuality, protecting “family honor”, economic pressure, unplanned pregnancy, cultural normalization, and the absence of educational and economic alternatives for girls. Healthcare and education professionals exhibited low preparedness to identify and intervene in cases.

2.3 Institutional Background

Historically, Brazilian criminal law established a direct and problematic relation between marriage, pregnancy, and sexual crimes. Under Law No. 2,848 of 1940, ([Brasil, 1940](#)), the punishability of certain sexual crimes – including rape, abduction, and seduction of minors – could be extinguished through marriage between the perpetrator and the victim. This provision remained in force for decades, effectively creating incentives for forced or coerced marriages, often in cases involving pregnancy. The institutional logic prioritized the “restoration of honor” and social legitimacy of unions over victim protection, thereby reinforcing patterns of child marriage and early motherhood.

This framework began to change with legal reforms in the last change of the century. Law No. 11,106/2005, ([Brasil, 2005](#)), abolished the possibility of extinguishing criminal liability through marriage, marking a critical shift away from the normalization of such practices. Subsequent reforms strengthened protections against sexual violence. Law No. 12,015/2009, ([Brasil, 2009](#)), expanded the legal definition of rape to include a broader range of sexual acts and introduced the category of “rape of a vulnerable person”, particularly protecting individuals under 14 years of age and those unable to provide consent. Later, Law No. 13,718/2018, ([Brasil, 2018](#)), clarified that consent is legally irrelevant in cases involving victims under 14, reinforcing strict liability and aligning the law with a more protective approach to minors.

Despite these advances, legal ambiguities remain, particularly regarding adolescents aged 14 to 18. While the Penal Code clearly defines individuals under 14 as legally incapable of consent, other provisions extend protections up to age 18 in cases of sexual exploitation. This inconsistency generates uncertainty about the validity of consent among older adolescents and may weaken enforcement, highlighting a persistent institutional gap.

Parallel to criminal law, the Statute of the Child and Adolescent (ECA), ([Brasil, 1990](#)), established the principle of “full protection”, recognizing children and adolescents as individuals in development with priority rights to health, education, dignity, and family life. Although the Statute of the Child and Adolescent does not explicitly prohibit child marriage, its framework implies that early unions are incompatible with the principle of integral development. Subsequent amendments expanded access to reproductive health services, psychosocial support for adolescent mothers, and preventive policies such as the National Week for the Prevention of Adolescent Pregnancy. These measures reflect a gradual change toward prevention and support rather than solely punitive approaches.

The institutional framework is further shaped by the National Council for the Rights of Children and Adolescents (Conanda), which issues binding resolutions guiding public policy ([Ministério dos Direitos Humanos e da Cidadania, 2026](#)). Recent resolutions explicitly clas-

sify child marriage and unions involving individuals under 16 as human rights violations and emphasize the need for comprehensive sexual education, access to reproductive rights, and protection against forced pregnancies. These guidelines reinforce the normative shift toward recognizing early marriage and adolescent pregnancy as multidimensional social problems requiring coordinated policy responses.

In addition to Penal Code, the Civil law also plays a central role in the phenomenon of child marriage. The Brazilian Civil Code, Law No. 10,406/2002, ([Brasil, 2002](#)), allows marriage from age 16 with parental consent, while historically permitting exceptions for individuals under 16 in cases of pregnancy or to avoid criminal penalties. These exceptions were eliminated by Law No. 13,811/2019, ([Brasil, 2019](#)), which prohibited marriage below age 16 without exception. However, remnants of the previous framework persist, including provisions that complicate the annulment of underage marriages – particularly when pregnancy is involved. These legal inconsistencies suggest that, despite formal prohibition, the institutional environment may still tolerate or fail to effectively deter such unions.

Overall, the evolution of Brazilian legislation reveals a gradual decoupling of marriage from the resolution of sexual crimes and a growing emphasis on the protection of minors' rights. Nevertheless, legal ambiguities and residual permissive provisions continue to create enforcement challenges, which may contribute to the persistence of child marriage and adolescent pregnancy.

Looking forward, recent legislative proposals reflect ongoing tensions in Brazil's institutional landscape. On one hand, some bills – PL n.º 195/2024 and PL n.º 728/2023 – aim to strengthen existing protections by closing legal contradictions – such as proposals to declare underage marriages null rather than merely voidable, and to extend age restrictions to informal unions, which are sometimes used to legitimize relationships involving minors ([Câmara dos Deputados, 2026](#)); ([Câmara dos Deputados, 2026](#)).

On the other hand, competing proposals – PL n.º 404/2021 and PL n.º 5011/2023 – seek to increase flexibility, for example by reducing parental consent requirements or allowing emancipated minors to marry without authorization. These initiatives often invoke adolescent autonomy, but raise concerns about weakening safeguards for vulnerable populations ([Câmara dos Deputados, 2026](#)); ([Câmara dos Deputados, 2026](#)).

A similar polarization emerges in debates on adolescent pregnancy and abortion. While some legislative efforts – PL n.º 3102/2025, PL n.º 4109/2024 and PL n.º 965/2022 –, focus on prevention, sexual education, and the recognition of new forms of sexual violence – e.g., non-consensual condom removal –, others propose stricter limitations on legal abortion, including equating late-term abortion with homicide even in cases of rape – PL n.º 1904/2024 – ([Câmara dos Deputados, 2026](#)); ([Câmara dos Deputados, 2026](#)); ([Câmara dos Deputados,](#)

2026); (Câmara dos Deputados, 2026). This divergence reflects broader ideological conflicts over reproductive rights, the protection of minors, and the role of the state.

From an economic perspective, such institutional inconsistencies can generate perverse incentives, reduce the effectiveness of public policies, and perpetuate cycles of vulnerability. The persistence of child marriage and adolescent pregnancy in Brazil thus reflects not only socioeconomic factors but also the incomplete alignment of legal institutions with their intended protective functions.

3 Model

To evaluate the mechanisms of child marriage in Brazil, we develop a discrete choice model in which a household – both the family and the child – chooses to marry or not, and if so in which margin of formality. We refer to the child as daughter as the literature for Brazilian demographics refers girls as the majority of the child in unions (Cardoso et al., 2022). Instead of using a matching model, as Zaki Wahhaj (2015), we choose a discrete choice model because the literature presents marriage not as a choice of a partner for the daughter, but as a mechanism to resolve previous relationships the girl had with an adult, being consensual or not (Taylor and Fonseca, 2015).

In this sense, we consider the girls’ age and if she is pregnant as state variables that will influence the choice of marriage. We also consider poverty a main driver for child marriage, therefore, both the household and the adult spouse wealth are determinants to their choice. Instead of considering a bride price or dowry, we describe the economic mechanism as an indirect wealth transfer from the adult spouse to the household as the girl marries. To understand social norms impact in this choice, we define parameters for the household weights to social norms, the general weight a region gives to marriage, and the reputational risk for the girls’ sexuality. We also adopt costs of legal punishment, which changes after 2019.

We distinguish households along two dimensions: the strength of economic pressure – primarily poverty and consumption constraints – and the strength of normative pressure – reputational concern, intergenerational transmission, social expectations. Table 1 presents the resulting typology.

Table 1: Mechanism typology of child marriage

		Low normative pressure	High normative pressure
High economic pressure	eco-pres-	Type A: Pure poverty. Income policy fully eliminates marriage.	Type B: Reinforced. Income policy partially reduces; norms sustain a residual.
Low economic pressure		Type D: Neither. Marriage unlikely.	Type C: Pure norm. Income policy has zero effect. Law may or may not bind.

A legal ban should, in principle, reach Types B and C if enforced, but leaves Type C households with strong substitution incentives toward informal unions. Types B and C are also the categories where post-pregnancy “repair” unions are common, since the pregnancy precedes the income or legal intervention.

We model the marriage-timing decision as a finite-horizon dynamic discrete-choice problem (DCDP) following the framework of [Rust \(1987\)](#). Time is discrete, each period corresponds to one year of the girl’s life. Let i refer to a household with daughter aged $a \in \{10, 11, \dots, 17\}$. In each period, the household and the daughter decides between to marry formally ($M = F$), marry informally ($M = I$), or wait ($M = W$). The state variable is a pair (a, p) , where $a \in \{10, \dots, 17\}$ is the age, and $p \in \{0, 1\}$ indicates pregnancy. A match offer is a pair $(\Delta, w(\Delta))$ drawn from market distribution $F_\Delta(a)$, where Δ is the age difference between the adult and the child of the couple and $w(\Delta)$ is the wealth contribution that the marriage represents to the household, which depends positively on the age gap.

The utility for each choice is as it follows:

$$U_i^F(a, p, \Delta) = w(\Delta) + \kappa_i N_i(\Delta) + \mu_i R(a) - \phi^F(a, p, t) + \varepsilon_{ia}^F \quad (1)$$

$$U_i^I(a, p, \Delta) = w(\Delta) + (\kappa_i - \lambda) N_i(\Delta) + \mu_i R(a) - \phi^I(a) + \varepsilon_{ia}^I \quad (2)$$

$$U_i^W(a, p) = Y_i + s(a) - \sigma p + \delta \mathbb{E}_{p'}[V(a + 1, p')] + \varepsilon_{ia}^W \quad (3)$$

The economic mechanism is driven by how attains the cost of the daughter’s needs and the future financial gain her education might bring. When a household decides to marry, formally or informally, there is cost relief for the household on the daughter’s costs that

is passed to the partner, which depends on the age gap of the couple. There is an indirect wealth transfer, $w(\Delta)$, from the adult of the couple to the family, which depends on how well established the adult is in the job market, therefore, the age gap, Δ , correlates positively with this indirect wealth transfer. When a household decides to postpone the marriage, they continue to have their own wealth Y_i , a possible gain with the girls' schooling, $s(a)$, and if the daughter is pregnant, a financial cost σ .

The social norms mechanism is translated into two variables: first, there is how much a household values marriage, $N_i(\Delta)$, which might depend on the age gap, and how much the family values social norms in general κ_i ; second, there is a reputational risk that depends on the daughter's age – this could mean the family values more marriage at a young age compared to marrying older, or could mean a risk of the daughter's reputation if she starts being sexual active –, therefore, this risk increases with age, and a family could give more or less weight based on μ_i . The variable λ is positive and represents a social cost of marrying informally.

Finally, the legal consequences are described by the function ϕ^F for formal marriages and ϕ^I for informal marriages. In the formal margin, before 2019, there was only a legal cost, $\bar{\phi}^F$, if there is no pregnancy, $p = 0$, and if the daughter is under sixteen, $a < \bar{a}_t$. Then, it depends on the judge to decide if the marriage can happen or if it transcribes to statutory rape. After 2019, the cost, $\bar{\phi}^F$, is certain if the girl is under sixteen, $a < \bar{a}_t$.

$$\phi^F(a, p, t) = \bar{\phi}^F \cdot \mathbf{1}[a < \bar{a}_t] \cdot [1 - (1 - \mathbf{1}_{t \geq 2019}) \cdot p] \quad (4)$$

For the informal margin, there is a cost, $\bar{\phi}^I$, of the union being reported to the legal authorities if the girl is under sixteen, $a < \bar{a}_t$.

$$\phi^I(a) = \bar{\phi}^I \cdot \mathbf{1}[a < \bar{a}_t] \quad (5)$$

Finally, if the household chooses to wait, there is the expected utility for the next period's decision with a discount factor δ . The expected value function satisfies:

$$V(a, p) = \mathbb{E}_\Delta \left[\log \left(e^{V^F(a, p)} + e^{V^I(a, p)} + e^{V^W(a, p)} \right) \right] \quad (6)$$

Where $V^j(a, p)$ is the deterministic part of the utility of choice j at state (a, p) , which includes the continuation value $\delta \mathbb{E}_{p'}[V(a + 1, p')]$ for the wait option. The expectation in (6) is over the match-offer distribution $F_\Delta(a)$, which we approximate by Monte Carlo integration using $N_{MC} = 300$ draws per age from the empirical distribution of Δ conditional on age in the PNADC subsample of in-union girls. The value function is solved by backward induction from $a = 18$ to $a = 10$.

We can first understand some theoretical implications of this model. The additive utility structure separates three distinct mechanisms – economic transfers, social norms, and legal deterrence – whose relative magnitudes are, in principle, separately identifiable. Because $w(\Delta)$ enters identically in U^F and U^I while the norm term enters with weight κ_i versus $\kappa_i - \lambda$, the choice between formal and informal marriage reveals the household’s valuation of social legitimacy relative to legal exposure, independent of the economic motive. Similarly, the margin between marrying and waiting reveals the net balance of the wealth transfer against the continuation value of schooling $s(a)$.

A central implication of the model is that making formal marriage more difficult to access does not necessarily reduce the total incidence of child marriage –it may simply displace households toward the informal margin. When $\phi^F(a, p, t)$ rises, as under Law 13,811 for all girls under sixteen after 2019, households compare the new cost of formal marriage against $\phi^I(a)$ and the social penalty λ of informality. If ϕ^I is substantially lower than ϕ^F and λ is small relative to $w(\Delta)$, the model predicts a compositional change, where fewer formal child marriages would occur and more unions would be informal, and implicating in a small reduction in overall marriage rates. This implies that measuring policy effectiveness solely through registered marriages could overstate the law’s impact on child welfare.

Because legal costs are discontinuous at $\bar{a}_t = 16$, the model generates a non-monotonic marriage hazard with respect to age. As the daughter approaches sixteen, the continuation value $\mathbb{E}_{p'}[V(a+1, p')]$ rises sharply for households exposed to high ϕ^F or ϕ^I , because waiting one additional period eliminates the legal penalty entirely. This creates a bunching prediction at age sixteen, where marriage rates should spike just at or after the legal threshold, and this increase should be more pronounced for the formal margin and attenuated among households with high κ_i or μ_i , who face stronger norm pressure not to delay.

Finally, the household-level heterogeneity in κ_i and μ_i – the weights on marriage norms and reputational risk – is what allows the model to distinguish economic from normative channels in a way that reduced-form regressions cannot. A household with low Y_i but low κ_i and μ_i will respond to an income transfer by reducing marriage rates, because the economic motive was the binding constraint. A household with moderate Y_i but high κ_i or μ_i will be largely unresponsive to income transfers but sensitive to interventions that shift perceived social penalties. The model predicts that the elasticity of child marriage with respect to cash transfers should be negatively correlated with the local prevalence of marriage norms.

4 Data

Our analysis draws on three complementary data sources covering the period 2003 to 2022.

Civil Registry Statistics (Registro Civil) Produced annually by the Brazilian Institute of Geography and Statistics (IBGE), the Civil Registry provides administrative records of all formally registered marriages in Brazil, disaggregated by the age and sex of each spouse, region, and year. We use marriages from 2003 to 2022 for all Brazilian municipalities. This source captures formal unions only, informal cohabitation and consensual unions are not reflected in these data.

National Continuous Household Sample Survey (PNADC) Conducted annually by the Brazilian Institute of Geography and Statistics (IBGE), the National Continuous Household Sample Survey is Brazil’s main household survey. We use annual waves from 2012 to 2019 and from 2022 to 2023, excluding 2020 and 2021 due to methodological adjustments during the COVID-19 pandemic. Survey weights are applied throughout using the calibrated person weight.

To identify unions involving minors, we restrict the sample to households containing both a self-declared household head and a spouse or partner of the opposite sex, and classify them into four categories: (A) both head and spouse under 18; (B) head under 18 and spouse 18 or older; (C) head 18 or older and spouse under 18; (D) both 18 or older. Categories A, B, and C together constitute our measure of early unions in the PNADC.

A key analytical choice involves comparing PNADC stock estimates of early unions to civil registry flow data. We construct a stock-based estimate from civil registry flows under the assumptions that (i) all unions involving minors persist at least until both members reach age 18 and (ii) marriages involving individuals younger than 15 involved 14-year-olds. This allows a consistent comparison and an estimate of the degree of underreporting.

5 Estimation

The estimation proceeds in two sequential stages. In the first stage we recover the legal deterrence cost $\hat{\phi}^F$ from a difference-in-differences design on administrative data. In the second stage, conditioning on $\hat{\phi}^F$, we recover the structural parameters governing utility through maximum likelihood on the individual PNADC panel. Both stages exploit distinct sources of variation: the first relies on aggregate panel variation across the age-16 legal threshold over time, while the second exploits cross-sectional heterogeneity in age, income, age gap, and pregnancy status across individual girls. Given the estimated parameters, we simulate counterfactual income and law-enforcement scenarios to quantify a norm irreducible share of child marriages.

The legal cost of formal marriage below age 16, $\bar{\phi}^F$, is not directly observed but can be

recovered from the behavioral response to the 2019 reform. Before March 2019, this cost was waived for pregnant girls, making $\phi^F = 0$ for the main constituency of below 16 formal marriages. After March 2019, these exceptions were banned, therefore $\phi^F = \bar{\phi}^F$ for all girls below 16 regardless of circumstances. The reform therefore creates a sharp change in the effective legal cost faced by one age group, less than 16, relative to an unaffected control group, 16 and 17, generating a difference-in-differences design.

We use the two-way fixed-effects on the number of marriages for age a at year t and regional state s , n_{ast} . Therefore, the estimative $\hat{\phi}^F$ quantifies the cost of marrying under 16 before 2019. We cluster standard errors by state.

$$\log(n_{ast} + 1) = \alpha_s + \alpha_a + \alpha_t + \hat{\phi}^F \cdot \mathbf{1}[a < 16] \cdot \mathbf{1}[t \geq 2019] + \varepsilon_{ast} \quad (7)$$

The coefficient on $\mathbf{1}[a < 16] \times \mathbf{1}[t \geq 2019]$ is sign-flipped when assigned to $\hat{\phi}^F$ so that a positive value denotes a deterrence effect: a reduction in the log count of formal marriages for the below-16 group relative to the 16–17 control group after the reform. The identifying assumption is that, absent the reform, the below-16 and 16 to 17 age groups would have followed parallel trends in formal marriage rates.

The model in Section 3 posits three choices: formal union, informal union, and waiting. However, PNADC does not separately identify the formal to informal nature of an ongoing union in a way that is consistent with the Registro Civil treatment: the household roster encodes whether a girl is living with a partner, not whether the arrangement is formally registered.

Absent a variable that separately records formal versus informal union status for the individual girl at the age of union formation, the choice-specific probabilities P^F and P^I cannot be separately identified from the available data. We therefore collapse $\{F, I\}$ to a single union alternative, yielding a binary choice set $\{\text{union}, \text{wait}\}$. This simplification preserves all the structural content of interest – the mechanism weights $(\alpha_0, \alpha_1, \mu, \gamma_0, \gamma_1)$ and the outside option parameters $(\beta_0, \beta_1, \sigma)$.

Given $\hat{\phi}^F$, the structural parameters are estimated by weighted maximum likelihood as defined below, where the outcome $y_i \in \{0, 1\}$ equals one if girl i is observed in union and zero if she is in the wait group. Survey calibration weights w_i are normalized to a mean of one. $\mathbb{P}_i$ is the probability of union, which is the logistic difference between the expected utility of union and expected utility of waiting, which are evaluated using the backward-induction solution to (6).

$$\begin{aligned}
V_i^{\text{union}} &= w(\Delta_i; \alpha_0, \alpha_1) + \mu R(a_i; \gamma_0, \gamma_1) \\
V_i^{\text{wait}} &= Y_i + s(a_i; \beta_0, \beta_1) - \sigma p_i + \delta \mathbb{E}_{p'_i}[V(a_i + 1, p'_i)]
\end{aligned}$$

Therefore, the weighted log-likelihood objective is:

$$\hat{\Theta} = \arg \max_{\Theta} \sum_i w_i \left[y_i \log \mathbb{P}_i(\Theta) + (1 - y_i) \log(1 - \mathbb{P}_i(\Theta)) \right] \quad (8)$$

The norm intercept parameter κ is collinear with α_0 and is fixed at zero; α_0 absorbs the combined entry intercept. The parameters μ , γ_1 , and σ are constrained to be positive via log-reparameterization: $\mu = \exp(\log \mu)$, $\gamma_1 = \exp(\log \gamma_1)$, $\sigma = \exp(\log \sigma)$. Optimization uses BFGS via with $\delta = 0.95$ calibrated.

Standard errors are computed from 200 bootstrap replications, i.i.d. row resampling of the PNADC sample, warm-started from the converged $\hat{\Theta}$. All 200 replications converged. The delta method is used to translate bootstrap standard errors on log-scale parameters to the natural scale.

6 Results

This section presents both descriptive evidence on child marriage in Brazil between 2003 and 2022 as well as equilibrium parameters before the marriage ban for people under 16 years old. We explore the difference between formal marriages, based on Civil Registry, and total marriages, based on National Household Sample Survey. Later, we use the model presented at Section 3 to estimate parameters for the child marriage choices before the marriage ban for under 16 years old. We also perform counterfactual scenarios with policies related to income, legal punishment and social norms education.

6.1 Formal Child Marriage

First, we discuss the evolution of child marriage in the formal margin between 2003 to 2022. The number of total marriages increased until 2015, followed by a sharp decline in 2020 due to the COVID-19 pandemic. Although partial recovery occurred in subsequent years, marriage levels in 2022 remain below pre-pandemic levels. In relation to child marriage, Figure 1 shows that marriages involving minors per 10,000 of them exhibit a clear downward trend throughout the period. Child marriage declined between 2004 to 2022 in approximately

72%: from 21 to 6 unions per 10,000 inhabitants.

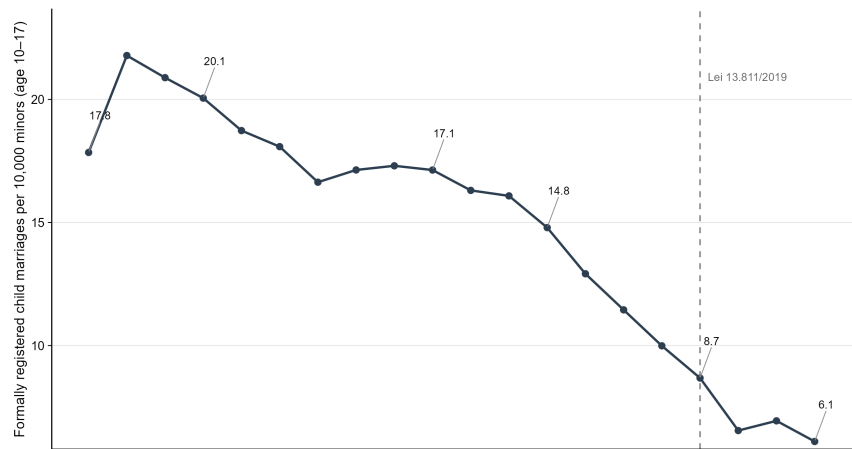


Figure 1: **Formally registered child marriages per 10,000 minors – ages 10 to 17 –, Brazil, from 2003 to 2022.** IBGE, Civil Registry Statistics. Numerator is formally registered marriages with at least one spouse under 18 in a given year, a flow measure. Denominator is the population estimates for the 10–17 age group. Dashed line marks Law 13.811/2019.

In Figure 2, we observe that in formal child marriages, women account for 91% of all minor spouses, a proportion that has remained remarkably stable, declining by only 4 percentage points over twenty years. Within the group of minor female spouses, 17-year-olds represent approximately 65% on average, and 16-year-olds about 25%.

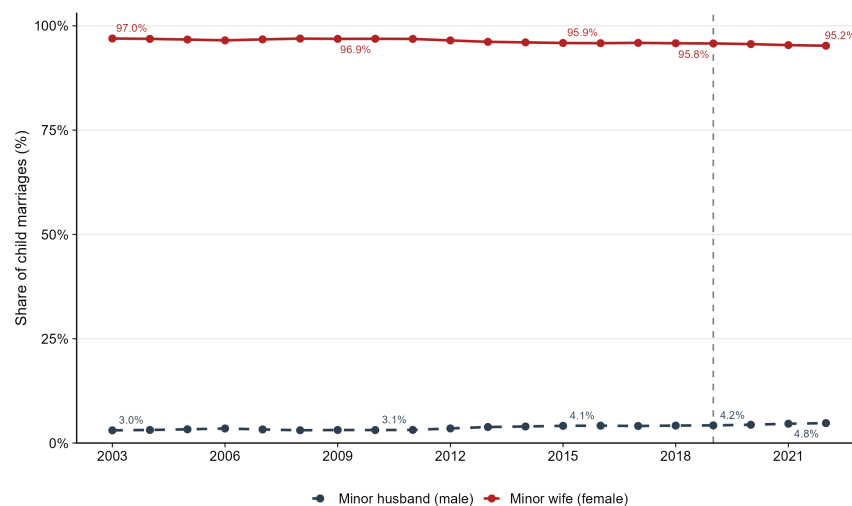


Figure 2: **Share of child marriages in the Civil Registry by sex of the minor spouse, from 2003 to 2022.** Source: IBGE, Civil Registry Statistics. Minor wife represents marriages in which the female spouse is under 18. Minor husband represents marriages in which the male spouse is under 18 and the female spouse is an adult. The two series sum to 100% within each year. Dashed line marks Law 13.811/2019.

We note that the downward trend in child marriage was well established before Law 13.811 took effect in 2019. Although, before 2019, formal child marriages continues to be officiated. Figure 3 presents the cumulative distribution of female child marriages to the age of the minor. Marriages involving girls under 16 persist, although most marriages before 17 decreased in the distribution.

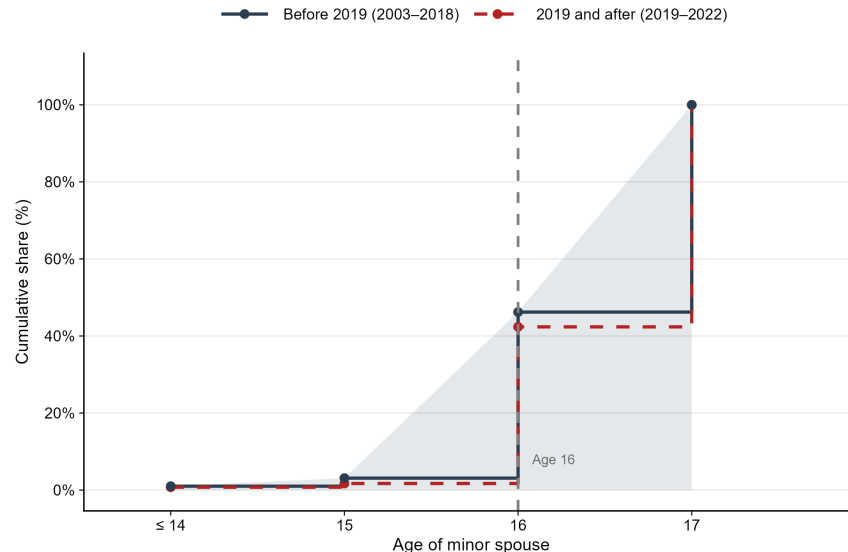


Figure 3: **Cumulative age distribution of the minor female spouse in formally registered child marriages: distribution before 2019 and after overlaying, restricted to marriages where the female spouse is the minor.** Source: IBGE, Civil Registry Statistics. A rightward shift in the post-2019 curve indicates a compositional change toward minor aged 17, following Law 13.811/2019.

The persistence of marriages below 16 years after 2019 represents that institutional representatives do not bind to the law: judges could still be using the exceptions to accept marriage for girls under 16 or agents at registry offices could still not fully understand the law change. In any case, the formal margin still absorbs marriages with spouses under 16, therefore, the increase in informal marriages is not necessarily expected.

The regional distribution of formal female child marriages is presented in Figure 4, where we show the proportion of female child marriages relative to marriages of women in their 20 to 29 years by Brazilian state, comparing 2003 and 2022. We use marriages between ages 20 to 29 for the greater incidence of marriage for women at this interval.

In 2003, states in North and North East, such as Rondônia, Alagoas, Paraíba, and Piauí consistently registered the highest proportions over the period, while São Paulo and Rio de Janeiro had the lowest proportions. Although virtually all states show a decline, the pace varies considerably, pointing to structural determinants, for example, Rondônia still marks

as highest rated of female child marriage, the only state to register more than 100 girls per 1,000 women as spouses in the formal margin.

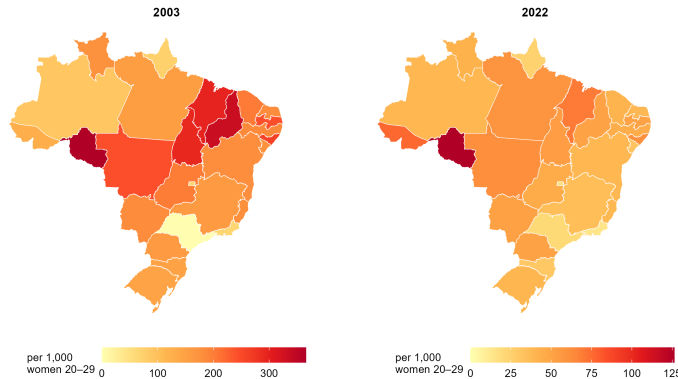


Figure 4: **Female child marriages per 1,000 marriages of women aged 20 to 29, by Brazilian state, 2003 and 2022.** Source: IBGE, Civil Registry Statistics.

Therefore, formal child marriage has a profile of being persistently represented by minors female spouses with adult men, frequently in poorer regions, with girls in ages 16 to 17 being more recurrent. Even after the Law 13.811/2019 banned exceptions to child marriage before 16 years of age, those marriages were still being officiated by formal institutions in Brazil.

6.2 Informal Child Marriage

One of the most significant contributions of this study to the current understanding of child marriage lies in the comparative analysis of the margins of formality through two data sources: the Civil Registry Statistics, which represents the formal unions officiated by registry offices, and the National Household Sample Survey (*PNAD Contínua* or PNADC), in which we can quantify all unions, including the informal margin.

To estimate the share of early unions that are never formally registered, we construct a synthetic stock of formal child marriages from Civil Registry (CR) flow data and compare it to the stock of early unions directly observed in the PNADC household survey. The CR records the annual flow of newly registered marriages involving at least one spouse under 18, disaggregated by the minor spouse’s age group. We convert this flow into a stock by recognizing that each registered marriage contributes to the count of active formal early unions for as many years as the minor spouse remains under 18: a marriage involving a 14-year-old bride contributes for four years, one involving a 15-year-old for three years, and so forth. The synthetic CR stock in a year is therefore the weighted sum of past flows whose

contribution window has not yet expired. This reconstruction requires no initial stock value, since the maximum contribution window is four years and CR data are available from 2003, nine years before the PNADC begins in 2012.

The PNADC stock, in turn, is the survey-weighted count of conjugal households in which at least one spouse is under 18, capturing both formal and informal unions. The underreporting rate in a year is then $1 - \frac{\text{CR stock}}{\text{PNADC stock}}$. Because the CR stock is a conservative lower bound – it counts only marriages that were formally registered and whose minor spouse has not yet aged out – the resulting underreporting estimate is itself a lower bound on the true extent of informal early unions.

As shown in Table 2, the results reveal substantial levels of underreporting in official records. The informal estimates are consistently between four and six times higher than formal data over the period analyzed. This corresponds to implied underreporting rates ranging from approximately 75% to 83%.

Year	Civil Registry	PNADC Survey	Ratio (PNADC / CR)	Implied underreporting
2012	71,002	292,844	4.1	75.8%
2013	67,215	266,575	4.0	74.8%
2014	64,299	279,426	4.3	77.0%
2015	59,441	272,652	4.6	78.2%
2016	52,017	247,506	4.8	79.0%
2017	45,043	235,947	5.2	80.9%
2018	38,438	210,913	5.5	81.8%
2019	32,700	199,862	6.1	83.6%

Table 2: **Comparison between officially registered unions involving minors and household survey estimates in Brazil.** The table contrasts the number of unions involving individuals younger than 18 years old recorded in the Civil Registry with estimates derived from PNADC household survey data. The ratio column reports the estimated number of unions in PNADC relative to officially registered unions, while the implied underreporting rate measures the share of unions not captured in the Civil Registry.

Moreover, the gap between the two sources increases over time. Figure 5 suggests that the discrepancy between formal records and the actual prevalence of unions involving minors has increased. In 2019, for instance, the average of daily unions involving minors was 91 per day in the formal unions, the general marriage data suggests a substantially higher figure of 550 daily unions.

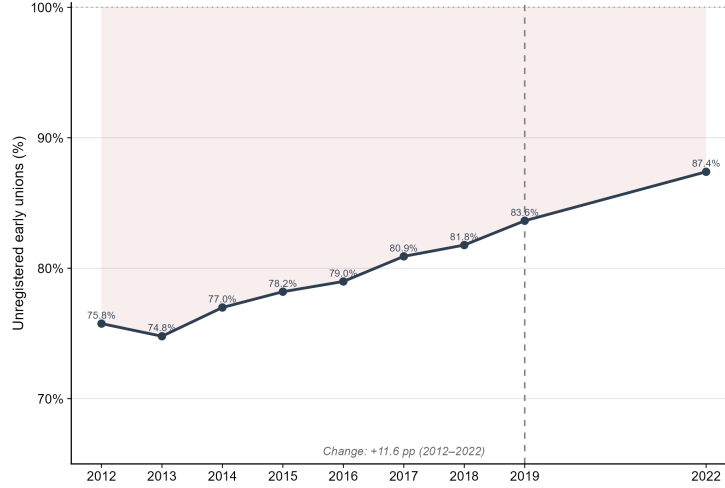


Figure 5: **Estimated underreporting of early unions, from 2012 to 2022.** Source: IBGE, Civil Registry Statistics and PNADC, first visit. The series shows the share of early unions that were never formally registered: $1 - \frac{\text{CR stock}}{\text{PNADC stock}}$. The CR synthetic stock is a weighted accumulation of past annual flows, where each marriage contributes for as many years as the minor spouse remains under 18. The PNADC stock captures all unions – formal and informal. Dashed vertical line marks Law 13.811/2019.

Table 3 presents the socioeconomic profile of household heads and spouses, disaggregated by the age composition of the household. There are three main demographic aspects of child marriage we emphasize.

First, households involving at least one minor are strongly overrepresented among Black and Brown individuals, with proportions ranging from 64 to 74%, compared to 52–54% in adult couple households. Second, there is a gender asymmetry in household roles: among households where the spouse is a minor (category C), 95.6% of those spouses are women, confirming that early marriage is disproportionately a female phenomenon. Third, income disparities are substantial: average monthly income in category A households – both under 18 – is approximately R\$750, compared to R\$2,550 in adult couple households. The domestic care burden is also notably higher for members of early-union households, with over 60% of individuals reporting domestic responsibilities or health constraints as the reason for not working.

We also observe the age gap between spouses: in households where the spouse is a minor (category C), the average age difference between head and spouse is approximately eight years. When both members are under 18 (category A), the gap is only one year – consistent with peer relationships – whereas category C households more closely resemble the pattern documented in the literature as a power asymmetry risk factor for intimate partner violence.

Understanding the average profile of child marriage, we can observe the change in distri-

	Household age composition			
	Both < 18	Head < 18 Spouse $\geq$ 18	Spouse < 18 Head $\geq$ 18	Both $\geq$ 18
Panel A: Household head				
Black or Brown (%)	73.8	69.6	70.6	54.0
Female (%)	28.6	79.2	4.4	28.8
Literate (%)	99.0	99.0	96.5	92.9
Paid employment (%)	67.6	29.2	82.2	67.7
Monthly income (R\$)	750	714	1,096	2,550
Weekly hours worked	39	37	42	42
Not working (%)	62.8	67.5	23.7	39.3
Panel B: Spouse or partner				
Black or Brown (%)	64.7	71.4	70.2	52.2
Female (%)	71.4	20.8	95.6	71.2
Literate (%)	99.6	98.1	98.9	93.5
Paid employment (%)	31.4	73.9	16.5	54.2
Monthly income (R\$)	760	1,166	630	2,137
Weekly hours worked	40	43	34	40
Mean age gap (years)	1.1	6.8	8.2	5.5
Not working (%)	54.7	68.6	66.9	59.6

Table 3: Average socioeconomic characteristics of household heads and spouses/partners by age composition of the couple. The table reports pooled means from PNADC data (2012–2019 and 2022–2023). Columns classify households according to whether the household head and spouse/partner are younger than 18 years old. Monetary values are reported in Brazilian reais (R\$).

bution after the ban of exceptions for marriage before 16 years in 2019. Figure 6 presents the distribution of child marriage by the age of the minor spouse, before and after 2019, when the legal exceptions for marriage before 16 years of age were removed.

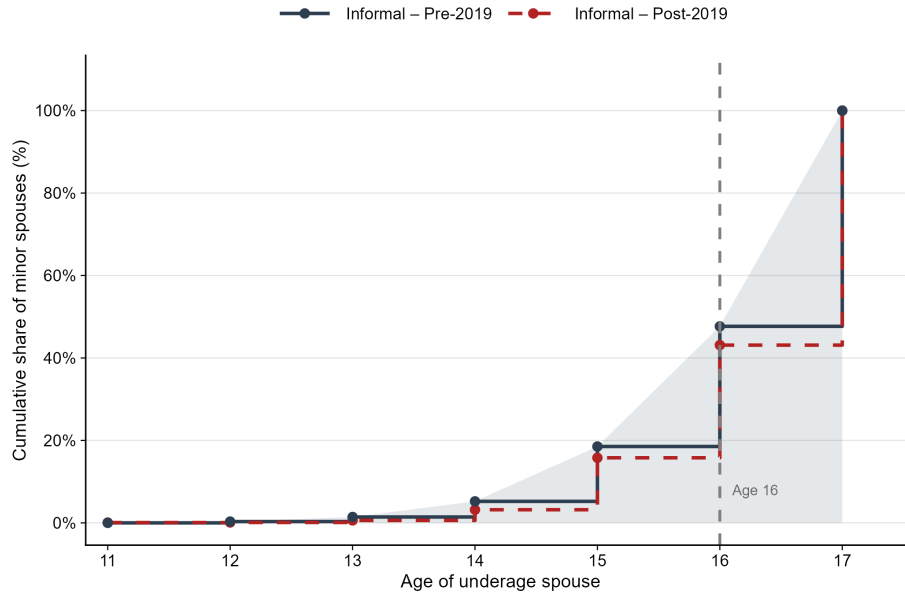


Figure 6: **Cumulative age distribution of minor spouses before and after 2019 for households in Category C in PNADC survey.** Source: IBGE, PNADC, first visit. Each curve shows the cumulative share of minor spouses aged at most x years, normalized within each period. Solid line marks the period before the legal change, from 2012 to 2018. Dashed line marks the period before legal change, from 2019, 2022 to 2023. The vertical dashed line marks age 16, the former exceptions threshold.

The figure indicates that child marriages were already concentrated among older adolescents prior to the reform, but became even more concentrated at ages 16 and 17 afterward, the same trend with formal marriages. Marriages involving children younger than 14 remained extremely rare in both periods, while the cumulative distribution begins to rise more substantially from age 15 onward. The largest increase occurs at age 16, although the cumulative share at this age declines slightly in the after 2019 period, suggesting a relative reduction in unions among younger minors. At the same time, the concentration at age 17 becomes more pronounced, which is consistent with a postponement effect induced by the reform. Overall, it suggests that the removal of legal exceptions did not eliminate child marriage in general, but may have shifted unions toward older ages immediately above the legal threshold.

6.3 Child Marriage Market

First, we present the estimates for equation 7 in Table 4.

	Dep. var.: $\log(n_{ast} + 1)$	
	(1)	(2)
$\mathbf{1}[a < 16] \times \mathbf{1}[t \geq 2019]$	0.411*** (0.022)	0.398*** (0.025)
State FE	✓	✓
Age-group FE	✓	✓
Year FE	✓	✓
State trends		✓
N	2,970	2,970
R^2	0.89	0.91

Table 4: *Note:* SEs clustered by state. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 5 presents the maximum likelihood estimates and bootstrap standard errors for the structural parameters.

Table 5: Structural DCDP parameter estimates

Parameter		Estimate	Boot. SE	95% CI	t
<i>Economic mechanism</i>					
α_0	Union entry (intercept)	−0.642	0.213	[−1.13, −0.18]	−3.0**
α_1	$w(\Delta)$ slope	−0.009	0.007	[−0.016, 0.010]	−1.3
<i>Norm and reputation mechanism</i>					
μ	Reputation weight	1.067	0.091	[0.89, 1.23]	0.76 [†]
γ_0	$\log R(a)$ intercept	−1.934	0.085	[−2.14, −1.83]	−22.7***
γ_1	$\log R(a)$ age slope	0.067	0.013	[0.047, 0.097]	5.1***
<i>Schooling and outside option</i>					
β_0	$s(a)$ intercept	−3.245	1.610	[−6.34, −0.34]	−2.0*
β_1	$s(a)$ slope in a	0.054	0.105	[−0.132, 0.257]	0.5
σ	Pregnancy cost	<i>unidentified</i> (see text)			—
<i>First-stage (Registro Civil DiD)</i>					
$\hat{\phi}^F$	Legal deterrence cost	0.411	0.022		18.7***

† log-scale t -stat; $^*p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. $N = 5,492$. $B = 200$ bootstrap replications.

Log-likelihood: -2,487.8. Discount factor $\delta = 0.95$ calibrated. κ fixed at 0 (collinear with α_0).

Notes: † t -statistic on log-scale parameter. $^*p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. $N = 5,492$; $B = 200$ bootstrap replications (100% convergence). Log-likelihood at $\hat{\Theta}$: -2,487.8. Discount factor $\delta = 0.95$ calibrated. κ fixed at 0 (collinear with α_0). Bootstrap CIs are percentile intervals.

6.4 Derived Economic Quantities and Mechanism Decomposition

The wealth-transfer payoff $w(\bar{\Delta}) = -0.72$ is negative, meaning that at the average age gap the economic component of union utility is a net cost. The insignificant slope $\hat{\alpha}_1$ suggests that the age gap alone is a poor proxy for $w(\Delta)$; partner income should enter directly in future specifications. Second, the union option nonetheless dominates waiting at all ages because $V^{\text{union}} > V^{\text{wait}}$, i.e., $-0.41 > -2.70$ at age 10 and $-0.23 > -2.32$ at age 17. The reputational mechanism fully offsets the economic deterrent. Third, the 59% increase in $R(a)$ between ages 10 and 17 implies that the reputational pressure to marry grows substantially with age, providing the primary structural explanation for the age gradient in marriage propensity that the data document.

7 Conclusion

We present the patterns of child marriage in Brazil over the past two decades, reaction to legal restrictions and mechanisms driving these outcomes. Overall, child marriage exhibit a consistent downward trend, although being consistently underreported and legal restrictions being disregarded.

We first note a pattern long recognized, child marriage remains strongly gendered: in 2022, girls accounted for approximately 91% of formal unions involving minors and tend to enter such unions at younger ages than boys. Despite the Law 13,811 banning the exception of pregnancy for marriage below 16, cases involving girls under this age restriction persist, indicating that the phenomenon has not been eradicated. In relative terms, unions involving minors declined by approximately 72% per 10,000 inhabitants in the period observed.

A key finding concerns the magnitude of underreporting. Comparative evidence from PNADC and Civil Registry data indicates that official records capture only a fraction of unions involving minors, with implied underreporting rates ranging from approximately 75% to 83%. These findings underscore the central role of informality in shaping both the measurement and the persistence of the phenomenon.

More broadly, the evidence highlights that early unions remain closely associated with socioeconomic vulnerability, including low income, limited educational attainment, and precarious labor market conditions, reinforcing cycles of inequality.

Despite the legal reforms introduced by Law 13,811/2019, which prohibited child marriage, these are still being officiated in the formal margin. This persistence points to a gap between formal legal frameworks and actual social practices. Moreover, recent legislative proposals and judicial decisions suggest the coexistence of institutional advances and potential setbacks, revealing tensions in the protection of children's and adolescents' rights.

As a next step, this research will move beyond descriptive analysis to conduct a causal impact evaluation of the 2019 legal reform. By exploiting the law as an exogenous institutional change, the objective is to identify its effects on the incidence of child marriage, adolescent fertility, and related socioeconomic outcomes, thereby contributing to a more rigorous assessment of policy effectiveness.

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